Ahmedabad
University

CSE623 Machine Learning: Theory and Practice

Topic:
Machine Learning Based Risk Parity Portfolio Construction for Financial Markets

Weekly Report 5

Submitted to: Prof. Mehul Raval
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1. **Overview of Project:**

Risk parity has emerged as a widely adopted portfolio construction framework that allocates capital by equalizing risk contributions across assets rather than maximizing expected returns. Popularized in institutional asset management as an "all weather" strategy, risk parity portfolios are designed to deliver stable performance across different market regimes. However, the effectiveness of risk parity critically depends on accurate estimation of asset volatilities and correlations.

Recent advances in machine learning and statistical learning theory provide tools for denoising and dynamically estimating covariance structures. Techniques such as shrinkage estimators, random matrix theory based eigenvalue filtering, and hierarchical clustering provide more robust representations of dependency structures (tail risks) for a multi-asset portfolio.

This project builds a risk parity portfolio using machine learning techniques. Using daily US financial assets data from 2015–2025, we construct long-short and long-only portfolios using PCA-based risk parity and Hierarchical Risk Parity (HRP). We evaluate their out-of-sample performance from 2020 to 2025 and compare them against the S&P 500 benchmark.

2. **Overview of Week 5:**

Tasks accomplished this week:

- Implemented Hierarchical Risk Parity (HRP) as an alternative allocation method to the PCA + GMV pipeline developed in Weeks 3–4. HRP uses unsupervised machine learning (hierarchical agglomerative clustering) to discover latent asset group structures and allocates risk through recursive bisection, requiring no matrix inversion.
- Built the complete HRP mathematical pipeline: correlation-based distance metric, Ward linkage hierarchical clustering, quasi-diagonalization (seriation), and recursive bisection with inverse-variance portfolio weights within each cluster.
- Ran the first HRP back-test using Marčenko–Pastur denoised covariance with the same rolling window framework (4-year training, 21-day holding) used in the PCA pipeline. Generated the equity curve and computed performance metrics.
- Reused all shared infrastructure from the PCA pipeline: data preparation (yield-to-price conversion, log returns), covariance estimators (raw, Ledoit–Wolf, Marčenko–Pastur), weight explosion guard, and performance evaluation framework.

3. **Work Completed:**

Correlation-Based Distance Metric

- HRP requires a distance measure rather than a similarity measure. The correlation matrix ρ (extracted from the denoised covariance via $\rho_{ij} = \Sigma_{ij} / \sqrt{(\Sigma_{ii} \cdot \Sigma_{jj})}$) was converted to a distance matrix using the formula $d_{ij} = \sqrt{((1 - \rho_{ij}) / 2)}$. This metric maps perfect correlation (ρ=1) to zero distance and perfect anti-correlation (ρ=−1) to unit distance, satisfying the triangle inequality required for valid clustering.

Hierarchical Agglomerative Clustering (Ward Linkage)

- Ward's minimum variance linkage was applied to the distance matrix. At each step, the algorithm merges the pair of clusters that minimizes the total within-cluster variance: $\Delta(A,B) = (n_a \cdot n^b)/(n_a+n^b) \cdot \|\mu_a-\mu^b\|^2$. Ward linkage was chosen over the single linkage method described in López de Prado's original 2016 paper because it produces balanced, compact clusters and avoids the "chaining problem" where single linkage creates elongated, unbalanced dendrograms. With 474 assets, balanced clusters are essential for the recursive bisection to achieve meaningful diversification.
- The clustering output is a linkage matrix Z of shape (N−1, 4), encoding the complete merge history as a binary tree (dendrogram). Each internal node represents one merge, and the N leaves correspond to individual assets.

Quasi-Diagonalization (Seriation)

- The dendrogram leaf ordering was used to reorder the covariance matrix so that correlated assets are adjacent. Mathematically, if π is the permutation from the leaf ordering, the quasi-diagonalized covariance is $\tilde{\Sigma} = P\Sigma\_MP\ P^T$. This creates a block-diagonal structure where large covariance values are concentrated along the diagonal, enabling the recursive bisection to split genuinely different risk groups at each step.

Recursive Bisection (Core HRP Allocation)

- The recursive bisection replaces the GMV formula $w = \Sigma^{-1}1 / (1^T\Sigma^{-1}1)$ from the PCA pipeline. It works top-down: starting with all assets weighted at 1, the sorted list is split into a Left branch (L) and Right branch (R) at each fork.
- For each cluster, inverse-variance portfolio (IVP) weights are computed using only diagonal elements: $w_i = (1/\Sigma_{ii}) / \Sigma(1/\Sigma_{jj})$. The cluster variance is then $V = w^T\Sigma w$ (including off-diagonal covariances). Capital is split inversely proportional to variance: $\alpha\_L = V\_R / (V\_L + V\_R)$, so the lower-risk cluster receives higher weight.
- This process repeats recursively down every branch until individual assets are reached. The final weights are guaranteed to be non-negative and sum to one — long-only by construction, with no matrix inversion required.

First HRP Back-Test (Marčenko–Pastur Denoised)

- The HRP back-test was executed using the identical rolling framework from Week 4: 4-year training window (1,008 trading days), 21-day holding period, Marčenko–Pastur denoised covariance. The weight explosion guard (1.25× threshold) was reused for consistency.
- At each rebalancing date, the pipeline computes the denoised covariance, extracts correlation, builds the dendrogram, quasi-diagonalizes, runs recursive bisection, and applies the guard. Portfolio PnL is computed over the subsequent 21-day forward window.

4. Challenges Encountered:

- Choosing between Ward and single linkage required careful analysis. López de Prado's original 2016 paper uses single linkage on a column-Euclidean distance matrix, but this suffers from the chaining problem with large N. We opted for Ward linkage based on its use in Riskfolio-Lib and subsequent literature, and documented the deviation.

- The recursive bisection splits the sorted list at the midpoint, which is optimal only when clusters are roughly equal in size. Ward linkage helps produce balanced clusters, but some rolling windows still produce uneven splits where one branch contains significantly more assets than the other.
- Integrating HRP into the existing codebase required ensuring that all shared functions (data preparation, covariance estimators, performance metrics) worked seamlessly with the new HRP weight computation, as HRP produces long-only weights whereas the PCA pipeline produces long-short weights.

5. **Plan for Week 6:**

- Run HRP with all three covariance estimators (raw sample, Ledoit–Wolf shrinkage, Marčenko–Pastur denoised) and compare their out-of-sample performance head-to-head.
- Analyze weight distributions across estimators to assess concentration and stability.
- Compute rolling 252-day correlation with the S&P 500 benchmark for each HRP variant.
- Generate visualizations including equity curve overlays and weight evolution plots.

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